Lorem Ipsum Dolor Sit Amet

Lorem Ipsum Dolor Sit Amet Consectetur
Lorem Ipsum University

Abstract

The task of motion planning is equivariant to rigid-body transformations, but the field that the flow matching network learns is not. In the current literature, there are three techniques to restore equivariance: in the data, in the operator, or in the weights.

A motion planning task names a start and a goal, and those two points already determine a canonical frame. No existing work measures what that frame is worth before any technique is applied, and none separates how equivariant a trained model became from how much that equivariance bought it.

We ask both. Holding architecture, data and training budget fixed, we build all three mechanisms on PointMass3D, a cluttered 3D benchmark, and score every model on problems it never saw, alongside a straight line from start to goal and three classical planners. The frame removes five of the six degrees of freedom of SE(3) exactly, at initialisation and at no cost, and is worth +36.5 points. The three mechanisms for the sixth are worth under a point each, at convergence, though the constrained architecture reaches any given level two to three times sooner, so the symmetry buys optimisation rather than accuracy. Two models that are equally equivariant differ by 28 points. We also report the measurement that limits our claim. Giving each waypoint its distance and direction to the nearest obstacle is worth more than the frame is, and the two are not substitutes: with that information supplied, the frame still adds +24.8 points.

I. Introduction

Motion planning is a fundamental problem in robotics, requiring the generation of collision-free, kinematically feasible trajectories from a start to a goal configuration in complex environments. Equivariance to rigid-body transformations is a cornerstone of generalization in this setting. It ensures that a planner solves structurally identical tasks identically, regardless of where in the world they are posed. Sampling-based [14] and optimization-based [24, 30] algorithms are equivariant by construction, since they act on the geometry directly, and they offer completeness or convergence guarantees. Their per-query computational cost limits real-time applicability. Learning-based planners [3, 8, 12, 22] amortize that cost across queries and have emerged as a promising alternative. They do not inherit the symmetry. A network trained on world coordinates relearns the same maneuver at every position and orientation.

Existing work restores it in one of three parts of the pipeline. In the data, through augmentation. In the inference operator, through frame averaging or canonicalization [13, 20]. Or in the weights, through architectures equivariant by construction [5, 6, 9, 26], as adopted in recent robot learning systems [23, 25, 28, 29]. All three modify the model. None measures how much of the symmetry the problem statement already supplies before any model is built.

A planning query names a start and a goal, and those two points determine a canonical frame. Place the origin at their midpoint, align the first axis with the direction between them, and express every waypoint and every obstacle in that frame. Three translational and two rotational degrees of freedom vanish exactly, at initialisation, at no computational cost and with no constraint on the architecture. The six numbers that conditioned the network on the query collapse to one invariant scalar. The frame needs a second axis to be determined, and ours takes it from a fixed world axis, so rotating the whole problem rolls the frame rather than leaving it unchanged. What survives is exactly that rotation: SE(3) collapses to SO(2). No continuous rule can fix the remainder [17], so the sixth degree of freedom must be handled by one of the three mechanisms above. Fig. 1 takes the group apart one stage at a time.

We build all three on a cluttered 3D benchmark, hold the architecture, data and training budget fixed, and score every model on problems it never saw, alongside a straight line from start to goal and three classical planners. The frame is worth +36.5 points of held-out collision-free rate. The three mechanisms for the sixth are worth +0.8, +0.68 and +0.15. A straight line from start to goal scores 15.6%, and the world-frame model, trained to a plateau, reaches 14.60%.

Our contributions are as follows.

- **Five of six degrees of freedom are free.** A frame derived from the query in closed form removes them exactly, raising the held-out collision-free rate from 14.60% to 51.10% with no change to the architecture, the data or the budget.
- **All three mechanisms for the sixth, measured against each other.** None recovers a point at convergence, but the constrained architecture reaches any given level two to three times sooner, so what the symmetry buys is optimisation rather than accuracy. We explain the null result rather than inferring it from flat curves, by measuring the non-

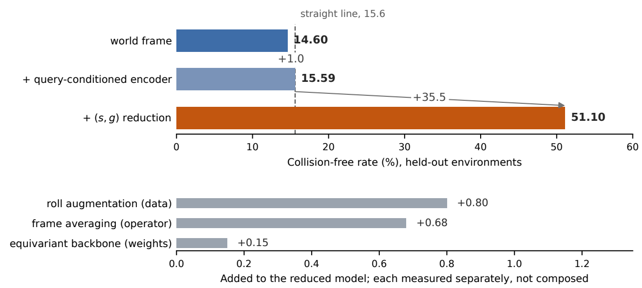

Figure 1: **Sixty environments, trained to convergence.** *Top*: a frame read off the query in closed form is worth +35.5 points over the bar immediately above it, and +36.5 over the plain world frame. Showing the obstacle encoder the same query without changing frame is worth +1.0. The dashed line is a straight segment from start to goal on the identical problems, which the world-frame model does not beat. *Bottom*: the three mechanisms for the one degree of freedom the frame cannot remove. Each is measured against its own baseline and they are alternatives rather than a decomposition, so they are not composed here and were never composed in the experiments.

equivariance residual of the learned field and proving it bounds what any post-hoc symmetrisation can remove. We then show the residual does not predict what a mechanism is worth: two models that are equally equivariant differ by 28 points.

- **Calibration in both directions, including against ourselves.** Supplying each waypoint with local geometry is worth +40.0 points where the frame is worth +36.5, so the representation change studied here is not the largest effect available on this benchmark. It is not a substitute for that information either: with local geometry present, the frame still adds +24.8.

II. Related Work

Classical planners. Sampling-based [14] and optimization-based [24, 30] planners act directly on geometry, so a rigid transformation of the problem transforms the solution. They offer probabilistic completeness or local convergence, at a per-query cost that follows from searching or optimizing at inference time. We use RRT-Connect and CHOMP to generate expert trajectories, and all three to bound what is achievable on our benchmark.

Learning-based planners. MPNet [22] and Motion Policy Networks [8] amortize planning into a network evaluated in milliseconds. Diffuser [12], Motion Planning Diffusion [3] and Diffusion Policy [4] generate whole trajectories with a denoising model, and flow matching [1, 15, 16] replaces the denoising chain with a learned velocity field, which is the objective we use. None of them inherits the symmetry that the classical planners get from the geometry for free.

Restoring the symmetry. The general framing is geometric [2], and equivariant models are known to carry a strict generalisation benefit under the right conditions [7]. Group-equivariant convolution [5], tensor field networks [26], SE(3)-Transformers [9] and vector neurons [6] build the constraint into the weights, packaged for practitioners by [10], and robot learning has adopted them widely [23, 25, 27, 28, 29]. Frame averaging [20] and learned canonicalization [13] act at inference instead, and the projection property our bound rests on is theirs. The Lie derivative [11] measures how equivariant a trained model became. That is the quantity we bound, and it is not the quantity, we find, that predicts anything.

III. Method

In this section, we begin by formally defining the generative motion planning objective and the symmetry it carries (Sec. III-A). We then decompose SE(3) into the part a planning query determines and the part it does not, and state precisely what survives that reduction (Sec. III-B). To establish that the remainder cannot be removed by the same route, we give two obstructions, one topological and one that applies to scenes carrying a symmetry of their own (Sec. III-C). Finally, we enumerate the three places a symmetry can be imposed, in the training data, in the inference operator, and in the weights (Sec. III-D), and define the non-equivariance residual that bounds what the second of them can remove.

III-A. Problem formulation

We formulate motion planning in the workspace $\mathbb{R}^3$. A task is the tuple $\mathcal{P} = (\mathbf{s}, \mathbf{g}, \mathcal{O})$, where $\mathbf{s}, \mathbf{g} \in \mathcal{C}$ are the start and goal configurations, $\mathcal{C} = \mathbb{R}^3$ for the point mass studied throughout and $\mathcal{C} = \mathrm{SE}(3)$ for the rigid-body domain of Sec. V-D, and $\mathcal{O}$ is a set of convex obstacles inducing a signed distance field $d_{\mathcal{O}} : \mathbb{R}^3 \to \mathbb{R}$. A trajectory is a waypoint sequence $x = [p_1, \ldots, p_N] \in \mathcal{C}^N$, and the feasible set of $\mathcal{P}$ is

$$\mathcal{S}(\mathcal{P}) = \left\{ x \ : \ p_1 = \mathbf{s},\ p_N = \mathbf{g},\ d_{\mathcal{O}}(p_i) \geq \rho \ \ \forall i \right\}, \qquad (1)$$

with ρ the robot radius. The objective is to learn a generative model $p_\theta(x \mid \mathcal{P})$ supported on $\mathcal{S}(\mathcal{P})$.

A rigid transformation $T \in \mathrm{SE}(3)$ acts on a task by $T\mathcal{P} = (T\mathbf{s}, T\mathbf{g}, T\mathcal{O})$ and on a trajectory waypointwise, $(Tx)_i = Tp_i$. The task is then exactly SE(3)-equivariant:

$$\mathcal{S}(T\mathcal{P}) = T\,\mathcal{S}(\mathcal{P}), \qquad \forall T \in \mathrm{SE}(3), \qquad (2)$$

and a generative planner should inherit this,

$$p_\theta(Tx \mid T\mathcal{P}) = p_\theta(x \mid \mathcal{P}), \qquad \forall T \in \mathrm{SE}(3). \qquad (3)$$

We use conditional flow matching [15], which learns a time-dependent velocity field $v_\theta(x_t, t, \mathcal{P})$ whose integral transports a simple prior to $p_\theta(x \mid \mathcal{P})$. Velocities are free vectors, so the corresponding condition on the field is

$$v_\theta(Tx_t, t, T\mathcal{P}) = R\, v_\theta(x_t, t, \mathcal{P}), \quad T = (R, \tau) \in \mathrm{SE}(3), \tag{4}$$

which yields Eq. (3) whenever the prior transported by the flow is itself invariant under the action. Nothing in the flow matching objective enforces either one, and Sec. V-E measures how far a trained field falls from Eq. (4).

III-B. Decomposing the group

The motion planning task gives us a canonical frame at initialisation, and at no cost. Unfortunately the canonicalization is not entirely complete, since fixing the rotation about the first axis needs a second reference axis, and the query does not give us one. We decompose as follows.

We declare $m = \frac{1}{2}(\mathbf{s} + \mathbf{g})$, $\ell = \|\mathbf{g} - \mathbf{s}\| > 0$ and $\hat{u} = (\mathbf{g} - \mathbf{s})/\ell$. The second axis is not determined by the query, and ours is read off the world basis vector e_k that $\hat{u}$ is least aligned with,

$$k = \arg\min_i |\hat{u}_i|, \qquad \hat{n} = \frac{\hat{u} \times e_k}{\|\hat{u} \times e_k\|}. \tag{5}$$

The denominator never vanishes: k forces $\hat{u}_k^2 \le \frac{1}{3}$, so $\|\hat{u} \times e_k\|^2 = 1 - \hat{u}_k^2 \ge \frac{2}{3}$. Let $R_\mathcal{P} \in \mathrm{SO}(3)$ be the rotation with $R_\mathcal{P}^\top = [\,\hat{u}\;\; \hat{n}\;\; \hat{u} \times \hat{n}\,]$. The reduction is the rigid map

$$T_\mathcal{P}(p) = R_\mathcal{P}\,(p - m), \tag{6}$$

under which the endpoints stop being inputs and become constants,

$$T_\mathcal{P}(\mathbf{s}) = \left(-\tfrac{\ell}{2},\, 0,\, 0\right), \qquad T_\mathcal{P}(\mathbf{g}) = \left(+\tfrac{\ell}{2},\, 0,\, 0\right). \tag{7}$$

Hence the network never sees a world coordinate. The task is carried into the frame, $\hat{x}_t = T_\mathcal{P}(x_t)$ and $\hat{\mathcal{O}} = T_\mathcal{P}\mathcal{O}$, obstacle centres transforming as points, box edges as free vectors under $R_\mathcal{P}$ alone, and radii not at all. The field is evaluated entirely inside the frame, $\hat{v} = v_\theta(\hat{x}_t, t \mid \ell, \hat{\mathcal{O}})$, where by Eq. (7) the query enters as the single scalar ℓ in place of the six numbers of $(\mathbf{s}, \mathbf{g})$. The output is returned to the world by the rotation alone, $v = R_\mathcal{P}^\top \hat{v}$, velocities being free vectors.

Eq. (6) costs one cross product per query, constrains nothing about v_θ, and holds at initialisation. It fixes five of the six degrees of freedom of SE(3), three of translation through m and two of rotation through $\hat{u}$. It does not fix the sixth, and Eq. (5) reads the world basis, so it turns with the world.

Proposition 1 (The residual is SO(2))**.** *Let $R_1(\theta)$ denote the rotation by θ about the first coordinate axis. For every task $\mathcal{P}$ with $\ell > 0$ and every $T \in \mathrm{SE}(3)$ there is a $\theta = \theta(T, \mathcal{P})$ with*

$$T_{T\mathcal{P}} \circ T \;=\; R_1(\theta) \circ T_\mathcal{P}. \tag{8}$$

Proof. $\|T\mathbf{g} - T\mathbf{s}\| = \ell$, so by Eq. (7) both $T_{T\mathcal{P}} \circ T$ and $T_\mathcal{P}$ carry $\mathbf{s}$ to $(-\frac{\ell}{2}, 0, 0)$ and $\mathbf{g}$ to $(+\frac{\ell}{2}, 0, 0)$. Their composite $A = T_{T\mathcal{P}} \circ T \circ T_\mathcal{P}^{-1}$ is therefore an orientation-preserving isometry fixing two distinct points, hence a rotation about the line joining them, which is the first coordinate axis. □

Nothing in the proof refers to Eq. (5). The residual is SO(2) for every rule selecting the second axis, and the rule decides only which θ. What Eq. (3) still asks of a planner in this frame is equivariance to SO(2), then, and not to nothing. One rule would escape it, a rule continuous in $\mathcal{P}$ and returning the same $\hat{n}$ for $\mathcal{P}$ and $T\mathcal{P}$. Ours is not continuous: it jumps where the minimising k in Eq. (5) ties. Sec. III-C is why no rule does better.

III-C. Why the remainder is not removable

The task of motion planning is equivariant to rigid-body transformations, but the field that the flow matching network learns is not. In the current literature, there are three techniques to restore equivariance: in the data, in the operator, or in the weights.

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Proposition 2 (Topological obstruction)**.** *There is no continuous $\hat{n} : S^2 \to S^2$ with $\hat{n}(\hat{u}) \perp \hat{u}$ for every $\hat{u} \in S^2$. Every rule selecting the second axis, Eq. (5) among them,* *is therefore discontinuous somewhere on the sphere of query directions.*

Proof. Such an $\hat{n}$ is a continuous nowhere-vanishing tangent vector field on S^2, which the hairy ball theorem forbids [17]. $\square$

Proposition 3 (Scenes with their own symmetry)**.** *Let $\hat{\mathcal{O}}$ be a reduced scene with $R_1(\pi)\,\hat{\mathcal{O}} = \hat{\mathcal{O}}$. Then no rule reading only the reduced task can distinguish the frame $R_{\mathcal{P}}$ from $R_1(\pi)\,R_{\mathcal{P}}$.*

Proof. $R_1(\pi)$ fixes both endpoints of Eq. (7) and fixes $\hat{\mathcal{O}}$ by hypothesis, so the two frames present the rule with identical arguments. $\square$

III-D. Three ways to impose the symmetry

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Averaging a field f over the K rolls $\theta_k = 2\pi k/K$ gives

$$(\mathcal{A}f)(\hat{x}) = \frac{1}{K}\sum_{k=0}^{K-1} R_1(\theta_k)^\top f\big(R_1(\theta_k)\,\hat{x}\big), \qquad (9)$$

Planner	Free (%)	Clearance	s/query
straight line	15.6	0.055	<0.001
TrajOpt	74.0	0.026	0.352
CHOMP	88.8	0.019	0.394
RRT-Connect	99.6	0.017	0.284
expert (RRT+CHOMP)	100.0	0.037	0.342
flow, world frame	14.6	−0.071	0.053
flow, $(\mathbf{s}, \mathbf{g})$ reduction	**51.1**	−0.002	0.063

Table 1: The same 500 unseen problems solved five other ways. The learned rows are converged, three-seed means at 60 environments. Classical timings are one CPU core and one query; the learned rows are one GPU and twenty samples, so the time column is not like-for-like. Clearance is negative when a path passes through an obstacle.

the orthogonal projection of f onto the fields equivariant to the cyclic subgroup $C_K \subset \mathrm{SO}(2)$ [20].

IV. Setup and Calibration

IV-A. Benchmark and protocol

PointMass3D places forty obstacles, twenty spheres and twenty oriented boxes, in $[-1, 1]^3$. The robot is a point of radius 0.03. Collision checking uses an analytic signed distance field, so it is exact rather than sampled. A trajectory is $N = 64$ waypoints in $\mathbb{R}^3$.

Expert paths come from RRT-Connect, then random shortcutting, uniform resampling, and CHOMP refinement. The dataset is 300 environments, each with 600 start-goal pairs and 30 paths per pair including time reversals. Environments 250 to 299 are held out and never trained on. Unless stated otherwise, models train on the first 60 of the remaining 250; the scaling study in Sec. V-A varies that count.

The planner is a conditional flow-matching model with 2.16M parameters: a dilated temporal convolutional trunk [18] with FiLM conditioning [19] and a PointNet-style obstacle encoder [21]. The encoder max-pools all forty obstacles into a single 128-dimensional vector, and FiLM applies that vector identically to every one of the 64 waypoints.

Every model is evaluated on the same 500 problems: 50 unseen environments, 10 distinct start-goal pairs each, 20 samples per pair, eight Euler steps. We report two metrics and never conflate them. A sample is *free* if the whole path is collision-free. *Best-of-20* asks whether any of the twenty samples for a query is free. They differ by about thirty points. Uncertainty is quoted across seeds for claims about a method. For comparisons against a fixed baseline measured on the same problems we use a paired bootstrap over environments.

IV-B. The floor, and a paired test against it

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V. Results

V-A. Scaling

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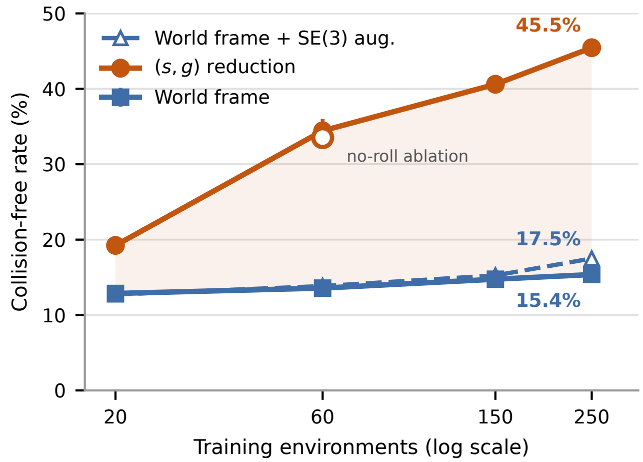

Figure 2: Lorem ipsum dolor sit amet consectetur adipiscing elit

Mechanism	20 ep.	conv.	
$(\mathbf{s}, \mathbf{g})$ reduction	+30.1	+36.5	n=3
3 translations	+12.3	—	recentring
2 rotations	+18.0	—	axis alignment
SE(3) augmentation	+2.1	—	7% of the above
roll augmentation	+0.8	—	n.s., n=3
frame averaging (K:1→3)	+0.1	+0.68	±0.13, n=3
equivariant backbone	—	+0.15	n=3

Table 2: Points on the held-out collision-free rate, each measured against the model without that mechanism. The rows are not additive and were never composed. The two degree-of-freedom rows were measured only on the fixed-budget grid, which is why that column exists.

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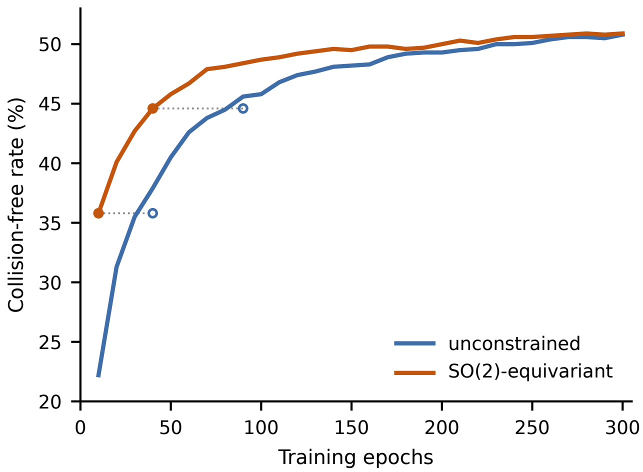

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V-B. Control: a query-conditioned encoder

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We ask both. Holding architecture, data and training budget fixed, we build all three mechanisms on PointMass3D, a cluttered 3D benchmark, and score every model on problems it never saw, alongside a straight line from start to goal and three classical planners. The frame removes five of the six degrees of freedom of SE(3) exactly, at initialisation and at no cost, and is worth +36.5 points. The three mechanisms for the sixth are worth under a point each, at convergence, though the constrained architecture reaches any given level two to three times sooner, so the symmetry buys optimisation rather than accuracy. Two models that are equally equivariant differ by 28 points. We also report the measurement that limits our claim. Giving each waypoint its distance and direction to the nearest obstacle is worth more than the frame is, and the two are not substitutes: with that information supplied, the frame still adds +24.8 points.

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V-C. Control: local geometry, and how it scales

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60 envs, converged	global only	+ local	geometry adds
world frame	14.60	54.54	+40.0
$(\mathbf{s}, \mathbf{g})$ reduction	51.10	79.37	+28.3
+ equivariant backbone	51.27	**82.13**	+30.9
the frame contributes	+36.5	+24.8	

Table 3: Local geometry against the query frame. Every cell is a mean over two or three seeds. The two local-geometry columns are the least seed-sensitive measurements in this work, with across-seed standard errors of 0.01 and 0.10 against 0.20 and 0.25 without.

equivariant differ by 28 points. We also report the measurement that limits our claim. Giving each waypoint its distance and direction to the nearest obstacle is worth more than the frame is, and the two are not substitutes: with that information supplied, the frame still adds +24.8 points.

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V-D. Replications

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The task of motion planning is equivariant to rigid-body transformations, but the field that the flow match-

Replication	Condition	World	Reduced	Gap
Second state space. SE(3) rigid body, floor 4.8				
	20 ep.	2.8	6.2	+3.4
	$\sim$156 ep.	3.1	**10.0**	+6.9
Other objective. DDPM, matched network evaluations				
	20 ep.	12.27	28.97	+16.71
	300 ep.	13.51	50.52	**+37.01**
Other clutter. Obstacles per environment				
	8	61.4	90.8	+29.4
	24	24.8	66.0	**+41.2**
	40 (trained)	13.5	46.4	+32.9

Table 4: The effect survives a change of state space, a change of generative objective, and a 5× range of clutter. The rigid body is an L-shaped union of five spheres whose state is a full pose, chosen to have trivial symmetry so the domain is not secretly easier; both models are below their own floor at 20 epochs and only the reduced one clears it. The density rows are freshly generated layouts at densities neither model was trained on. Single seed except the DDPM 20-epoch row.

ing network learns is not. In the current literature, there are three techniques to restore equivariance: in the data, in the operator, or in the weights.

V-E. The residual, and what it bounds

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Corollary 1 (Lorem ipsum)**.** *Lorem ipsum dolor sit amet consectetur adipiscing elit, sed do eiusmod tempor incididunt* $\|f - \mathcal{A}f\|^2 \leq r^2$.

VI. Limitations

- **The group must act on the state space, not just the workspace.** This holds for a point mass and for the rigid body of Table 4. It fails for a manipulator planning in joint space: the query still defines a frame in the workspace, but a rigid motion of the world induces no clean action on joint angles. Mobile bases, free-flying bodies and end-effector-space planning satisfy the condition. Joint-space planning for arms does not.
- **The learned planner loses to the classical ones.** RRT-Connect solves 99.6% in under 0.3 s on one CPU core. On this benchmark one should use RRT-Connect. What we measure is what a representation change is worth, and that is only isolable where the confounds are removable.
- **One workspace family, one clutter generator, one architecture.** The claim that the sixth degree of freedom needs no treatment relies on the training distribution supplying implicit roll diversity through many start-goal directions. A dataset with few directions per environment could reverse it.

- **Seeds are not uniform.** The 60-environment cells carry three seeds, the local-geometry cells two or three, and the 250-environment cells one. Every claim we lean on has at least two.
- **Local geometry assumes explicit primitives at inference.** The signed distance is a deterministic function of obstacle data the network already receives, so it is a featurisation rather than an oracle, but it is unavailable to a planner working from perception.

VII. Conclusion

Five of the six degrees of freedom of SE(3) were removable exactly, at initialisation and at no cost, by a frame the planning query already supplied. That was worth +36.5 points of held-out collision-free rate, against a world-frame model that did not beat a straight line. The sixth was not removable by any continuous rule, and all three mechanisms available for it recovered under a point each. The constrained architecture was not idle: it reached any given level two to three times sooner, so the symmetry bought optimisation and not accuracy. Two models that were equally equivariant differed by 28 points, so how equivariant a model became did not predict what the mechanism was worth. Giving each waypoint its distance and direction to the nearest obstacle was worth more than the frame was, and the frame still added +24.8 points on top of it.

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The software implementation was carried out with Claude Code under the first author's direction. The classical planner implementations, used to generate expert trajectories and to bound what is achievable, follow the published algorithms [14, 24, 30] and their public reference implementations. The problem formulation, the theoretical analysis, the experimental design and the text of this paper are the authors' own. Every reported number is produced by the released code, whose geometric and equivariance-critical components are verified by property-based tests that run on untrained networks and include negative controls.

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